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# Seismic data interpolation by frequency domain anti-aliasing greedy Radon transform

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## Summary

I present a new interpolation method using a frequency domain anti-aliasing greedy Radon transform. Based on the prior information of unaliased low frequencies, the method designs sparse weights to penalize spurious events due to aliasing and solves for optimal Radon model elements in a priorized way. A greedy algorithm is used to reduce the compuational cost and enhance signal-to-noise ratio. Synthetic and real data examples show that the algorithm can handle structural aliasing. By limiting the range of structural dips, the method naturally eliminates residual linear noise.

## Introduction

Due to financial budget and the limitation of surface conditions, seismic data is seldom acquired in a complete or regular way. As a remedy to this spatial sampling problem, seismic interpolation has been widely developed in the past two decades. In particular, a Minimum Weighted Norm Interpolation (MWNI) (Liu and Sacchi, 2004; Trad, 2009) using Fourier transform based inversion has become a popular solution to reduce migration artifacts thanks to better data sampling after high dimensional reconstruction. The method can effectively regularize/interpolate randomly sampled data. However, it cannot handle regularly missing and aliased data. Ng et al (2015) proposed an angularly weighted MWNI to reduce spatial aliasing in seismic data. The method computes weights based on slant stacking along the radial directions using all frequencies, which can attenuate spectrum replicas due to regularly missing data. Combining with the iteratively updated sparse weights, the improved MWNI can handle seismic data that has both irregular and regular sampling problems. In this paper, I propose an alternative solution using high dimensional Radon transforms. The new method can perform interpolation and noise attenuation in the same time. Both random noise and linear noise can be reduced due to the benefits of Radon transforms. Synthetic studies show that the new method is more stable than Fourier transform based methods when there is strong spatial aliasing in seismic data.

## Theory

I start from a 2D case for easy understanding of how to interpolate data using Radon transforms. In the space and temporal frequency (FX) domain, noise-free seismic data can be modeled using the following equation:

$$d(\omega, x) = \sum_p m(\omega, p) e^{i\omega p x}, \quad (1)$$

where $\omega$ is the angular frequency, $x$ is the spatial coordinate, $p$ is the moveout parameter, $d$ is the data in the space and temporal frequency domain, and $m$ is the Radon model in frequency and moveout parameter domain.

The adjoint operator of Equation (1) is

$$m(\omega, p) = \sum_x d(\omega, x) e^{-i\omega p x}. \quad (2)$$

Equation (2) describes a slant-stacking process in the FX domain to transform seismic data to a Radon model. The forward and adjoint operators can be described in matrix form as below:

$$d = Lm, \quad (3)$$
$$m = L'd. \quad (4)$$

Equation (1) shows that we can predict seismic data in any location if we know the Radon model. In practice, I try to derive an optimal model from limited observations by minimizing the following cost function

$$J(m) = ||d - Lm||^2 + \mu R(m), \quad (5)$$

where $R$ is a regularization function of the Radon model. The first term of the above cost function is used to fit input data, and the second term is used to force the Radon model to have specific features to avoid fitting the noise. People often choose L1 norm or Cauchy norm (Sacchi and Ulrych, 1995) to promote sparseness in the model. In this paper I utilize a priorized L1 norm to constrain the solution. The problem can be solved using a greedy approach (Wang 2021).

Once the optimal Radon model is computed by minimizing Function (5), we can predict the data at a new position using the following equation

$$d(\omega, x_{new}) = \sum_p m(\omega, p) e^{i\omega p x_{new}}. \quad (6)$$

When there is spatial aliasing in the data, the method using Function (5) is not enough to recover regularly missing data. In this paper I propose another sparse constraint designed with lower frequencies to penalize artifacts cost by spatial aliasing. Function (5) is recast as

$$J(m) = ||d - Lm||^2 + \mu R(Am), \quad (7)$$

where $A$ is a diagonal weighting matrix. The weight for each element of the Radon model is simply computed by summing the amplitude of corresponding model of all lower frequencies than current frequency to solve. Below is a greedy implementation of aniti-aliasing 2D Radon transform:

1. Convert data from $d(x,t)$ to $d(x,\omega)$ by 1D Fourier transform along the time direction. Intialize the amplitude weights $A(p)$ with zeros.
2. Loop over frequencies to find an optimal Radon model for each frequency using the following algorithm in pseudo code.

$d_{resi} = d, \quad m = 0;$
$m_e = L'd_{resi};$
$Update\ A(p)$ with $m_e$ by $A(p) = A(p) + A(m_e);$
$for\ iter = 1, nitr$
$\quad m_e = L'd_{resi};$
$\quad sort\ \ m_e\ \ in\ amplitude\ descending\ order\ based\ on\ weighted\ amplitude\ A(p).A(m_e) and$

*keep the sorting indices of first N samples ;*
*for i=1,N*
*optimize the weights of the i-th model element using the steepest descent method;*
*update the i-th model with the optimal model element;*
*update the data residual $d_{resi}$ by sutracting the predicted data using the optimal model;*
*end*
*end*

For 4D interpolation, I replace the modeling equation with the following

$$d(\omega, x, y, h) = \sum_{p_x, p_y, p_h} m(\omega, p_x, p_y, p_h) e^{i\omega(p_x x + p_y y + p_h h^2)}, \tag{8}$$

where *x* and *y* are two spatial coordinates, $h$ is the offset, $p_x$ and $p_y$ are slopes (normalized moveout parameters) in crossline and inline directions, $p_h$ is the slope in the offset direction, $d(x, y, h, \omega)$ is seismic data in the FX domain, $m(p_x, p_y, p_h, \omega)$ is the Radon model in the p and temporal frequency domain,

and the ajoint equation becomes

$$m(\omega, p_x, p_y, p_h) = \sum_{x,y,h} d(\omega, x, y, h) e^{-i\omega(p_x x + p_y y + p_h h^2)}. \tag{9}$$

In practice, seismic data is separated into several azimuth sectors, and then 4D interpolation is performed for each azimuth sector.

For 5D interpolation, the modeling equation is formulated as below

$$d(\omega, x, y, g, l) = \sum_{p_x, p_y, p_g, p_l} m(\omega, p_x, p_y, p_g, p_l) e^{i\omega(p_x x + p_y y + p_g g^2 + p_l l^2)}, \tag{10}$$

where $g$ is x-offset, $l$ is y-offset, $p_g$ is the slope in the x-offset direction, and $p_l$ is the slope in the y-offset direction. The adjoin equation for Equation (10) is

$$m(\omega, p_x, p_y, p_g, p_l) = \sum_{x,y,g,l} d(\omega, x, y, g, l) e^{-i\omega(p_x x + p_y y + p_g g^2 + p_l l^2)}. \tag{11}$$

It is straigtforward to replace corresponding operators in the previously described greedy implementaion of 2D Radon transform to perform 4D/5D Radon inversion and then use the optimized Radon model to reconstruct seismic data.

## Synthetic examples

To examine the performance of the proposed greey Radon interpolation, I prepare three sets of synthetic data, one with randomly missing traces and two with regularly missing traces. Figure 1 shows the result of the first dataset using greedy Radon transform. It can be seen that the algorithm successfully recovers all missing traces, and the difference between original data and the interpolated traces is ignorable. Note that in this example, no anti-alasing weights are required to obtain a satisfactory result.

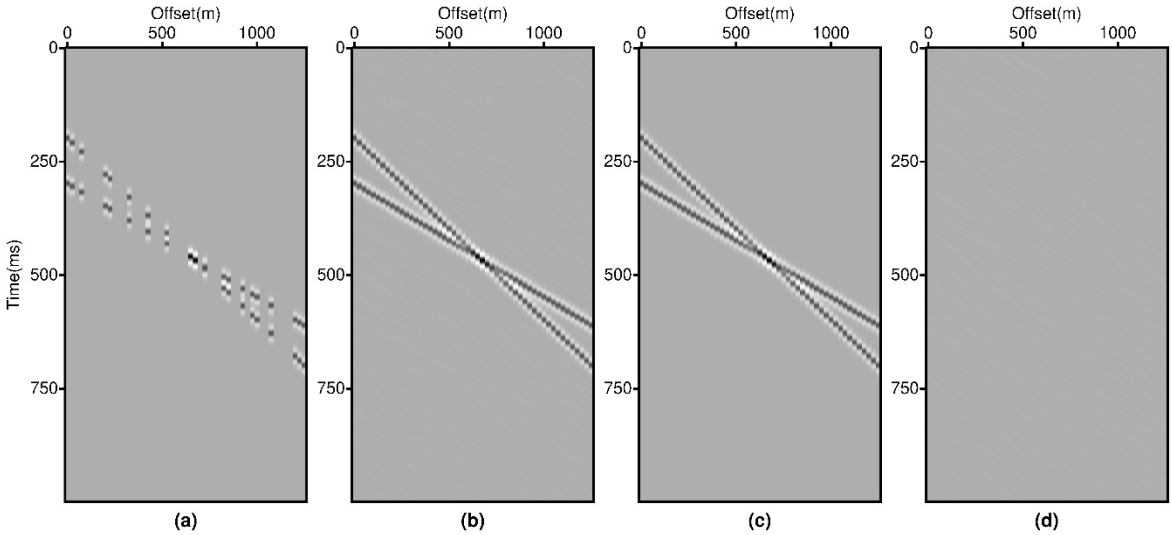



Figure 1: Test of greedy Radon interpolation on randomly sampled data. (a) Input data. (b) Interpolated data. (c) Original complete data. (d) Difference between original data and interpolated data.

Figure 2 compares the amplitude of adjoint Radon model (obtained by applying Equation (2) to input data) and the final Radon model (obtained by inversion). In Figure 2a, the missing traces appear as weaker and less organized events in the frequency and moveout (FP) domain than the true complete signal. As described further above, the greedy algorithm tries to identify the strongest event and optimize the weight for this event. Then contribution of the strongest event is computed and subtracted from input. Naturally the data leakage to false events is suppressed. As a result the model is updated with events in amplitude decreasing order until the data residual is small enough. Usually it only takes 4-6 iterations to obtain a satisfactory result. In Figure 2 it is clear that the greedy inversion method has greatly enhanced the resolution of Radon model.

Figure 3 shows the interpolation result on a dataset with regularly missing traces. Note that when all seismic events have the same dipping direction, conventional greedy Radon transform without anti-aliasing weights can work well as shown in Figure 3. However careful examination of the interpolated traces shows that some weak artifacts are introduced during the inversion.

Figure 4 demonstrates the advantage of anti-aliasing greedy Radon transform over conventional greedy Radon transform. The interpolated traces look cleaner than those in Figure 3. The comparison of amplitude of Radon models in Figure 5 confirms that with anti-aliasing implementation, the greedy Radon transform can suppress spurious aliasing events and improve the interpolation quality.

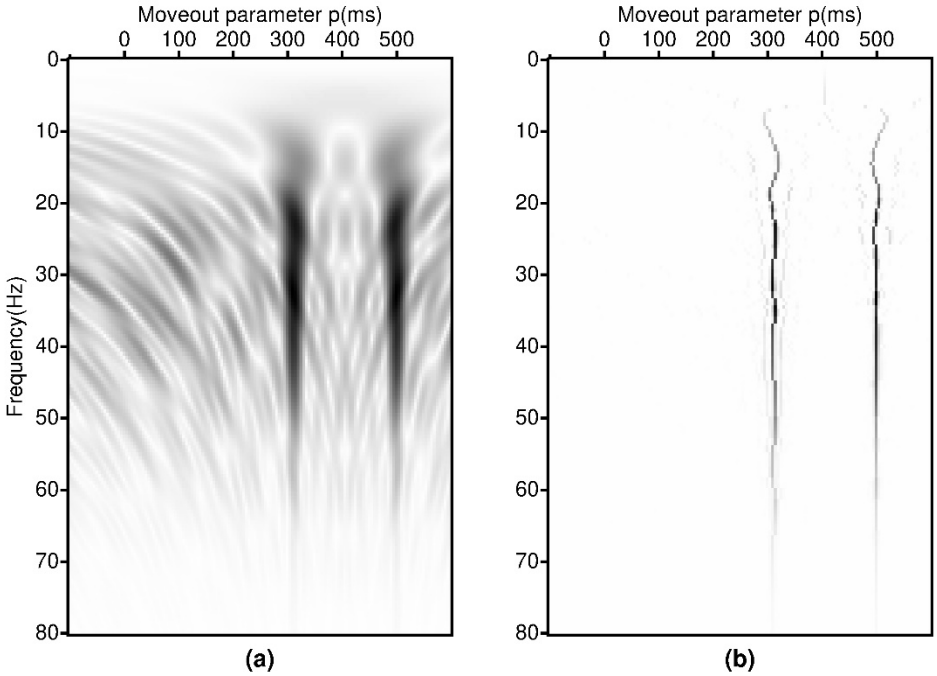



Figure 2: Comparison of the amplitude of adjoint Radon model and the inverted Radon model in the FP domain. (a) Amplitude of the adjoint Radon model. (b) Amplitude of the greedy Radon model.

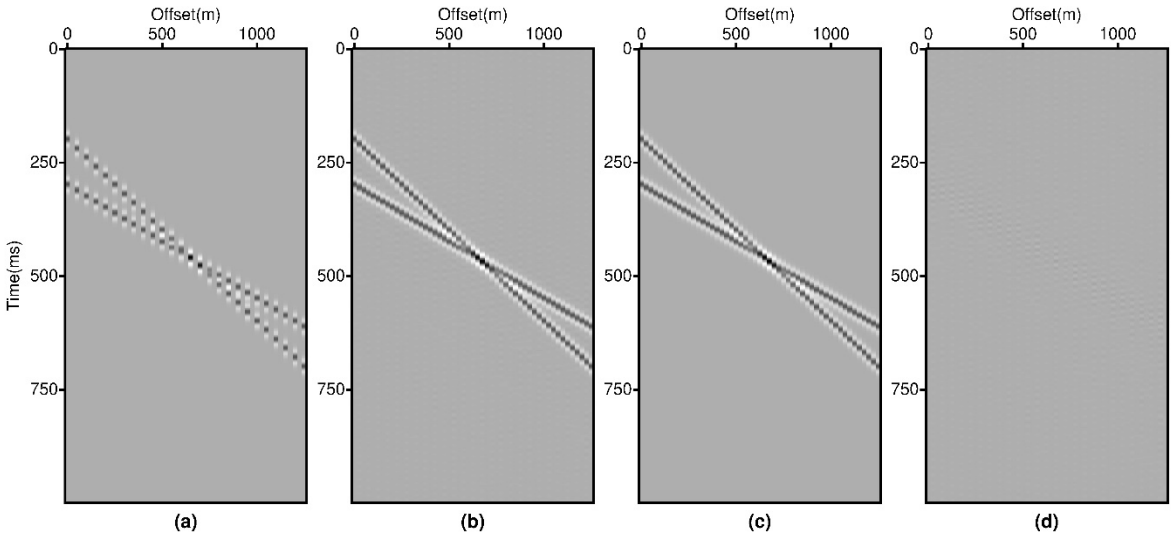



Figure 3: Test of greedy Radon interpolation on a dataset with regularly missing traces. (a) Input data. (b) Interpolated data. (c) Original complete data. (d) Difference between original data and interpolated data. Some weak artifacts are present at the locations of missing traces.

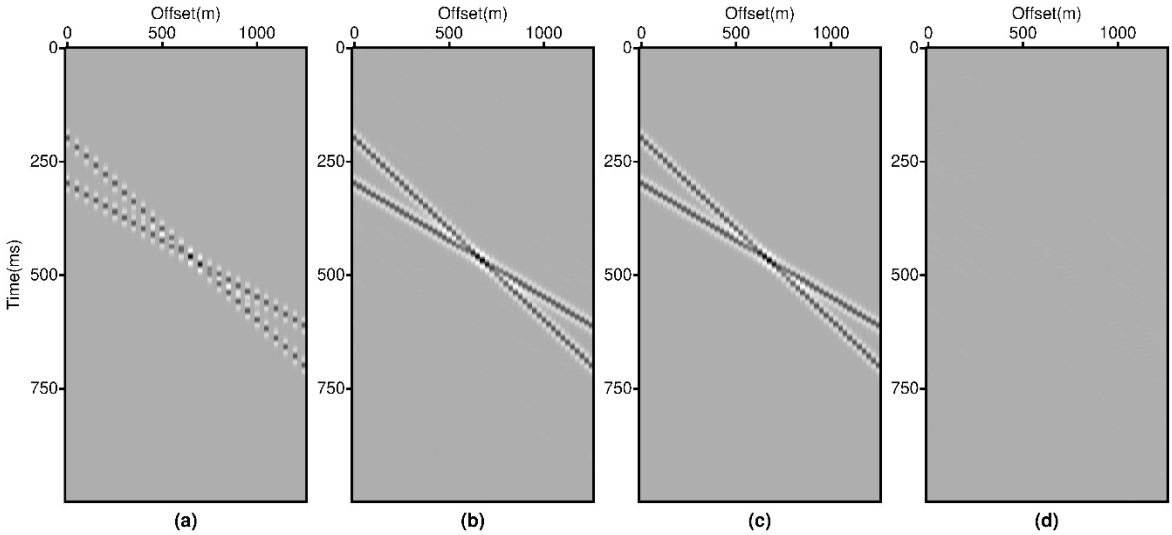



Figure 4: Test of anti-aliasing greedy Radon interpolation on a dataset with regulary missing traces. (a) Input data. (b) Interpolated data. (c) Original complete data. (d) Difference between original data and interpolated data. No obvious artifacts are present in the difference.

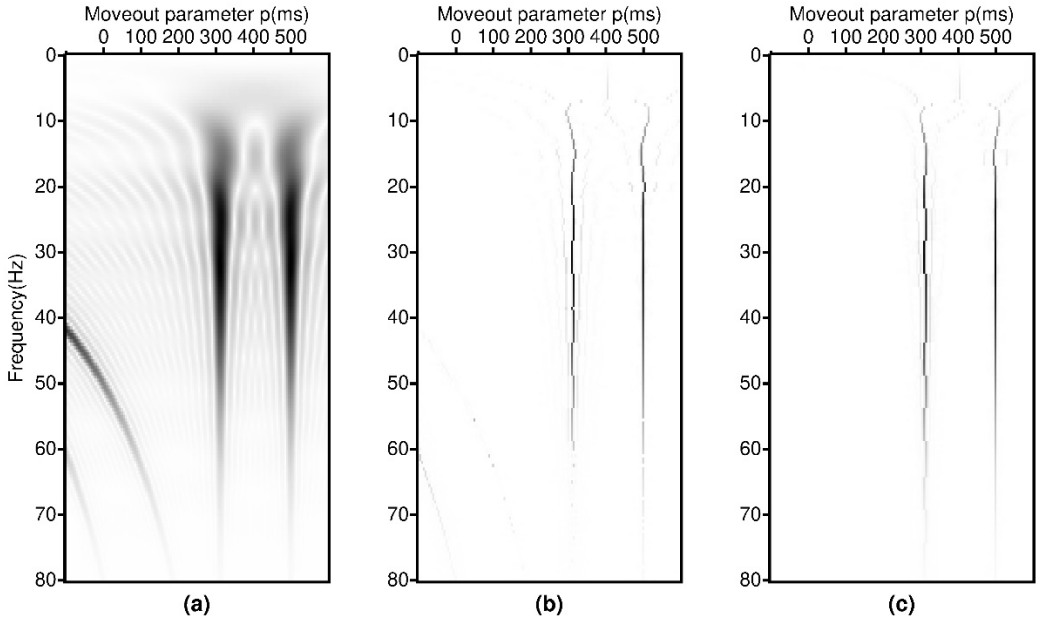



Figure 5: Comparison of the amplitude of adjoint Radon model and the inverted Radon models in the FP (frequency and moveout) domain. (a) Amplitude of the adjoint Radon model. (b) Amplitude of the greedy Radon model. (c) Amplitude of the anti-aliasing greedy Radon model.

When seismic data contains events with opposite dipping events, the benefits of the anti-aliasing addon are more evident. Figure 6 is the result of conventional greedy Radon transform on a dataset with regularly missing traces and there are two events with opposite dipping angles. It can be seen that the algorithm fails to recover regularly missing traces. The result looks even worse than the input because a lot of artifacts are introduced as shown in the difference plot (Figure 6d).

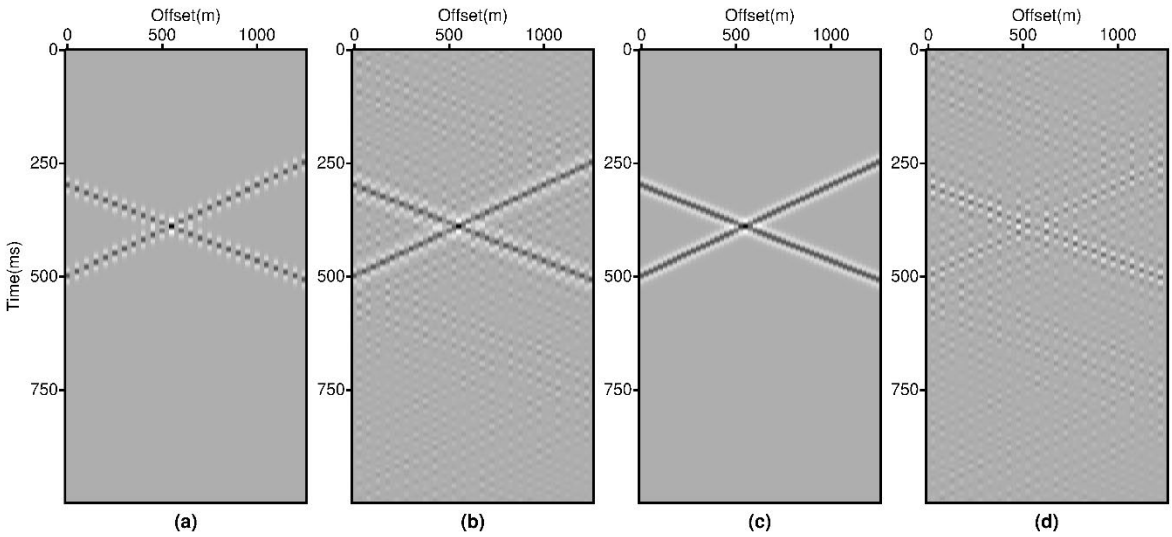



Figure 6: Test of greedy Radon interpolation on a dataset with regulary missing traces and events with opposite dipping angles. (a) Input data. (b) Interpolated data. (c) Original complete data. (d) Difference between original data and interpolated data.

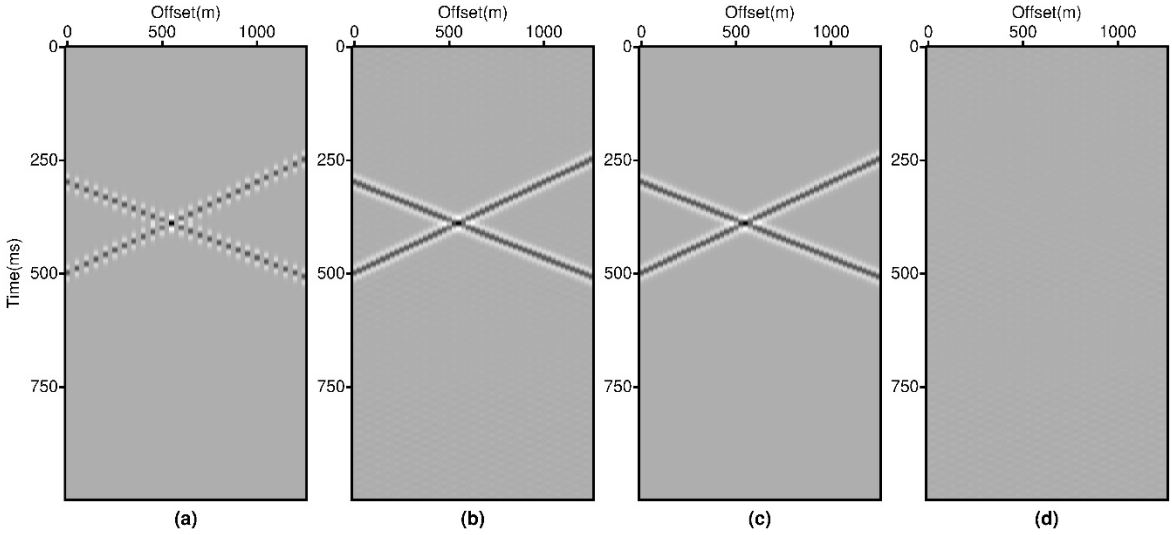



Figure 7: Test of anti-aliasing greedy Radon interpolation on a dataset with regulary missing traces and events with opposite dipping angles. (a) Input data. (b) Interpolated data. (c) Original complete data. (d) Difference between original data and interpolated data.

By contrast, the anti-aliasing version of the greedy Radon transform can handle this kind of situation. Figure 7 shows that all missing traces are successfully recovered by the new algorithm. In addition, the difference plot (Figure 7d) indicates that there are no severe artifacts present in the result.

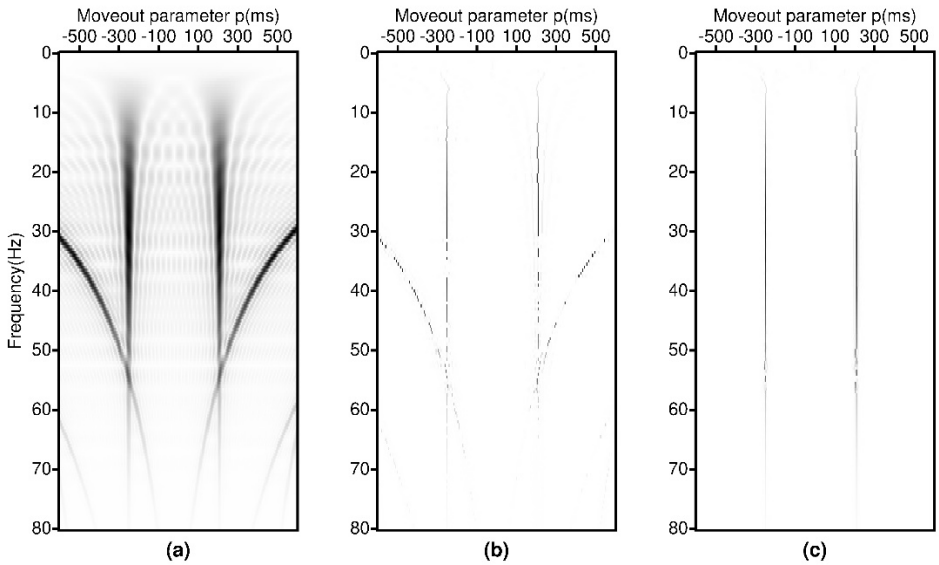



Figure 8: Comparison of the amplitude of adjoint Radon model and the inverted Radon models in the FP (frequency and moveout) domain. (a) Amplitude of the adjoint Radon model. (b) Amplitude of the greedy Radon model. (c) Amplitude of the anti-aliasing greedy Radon model.

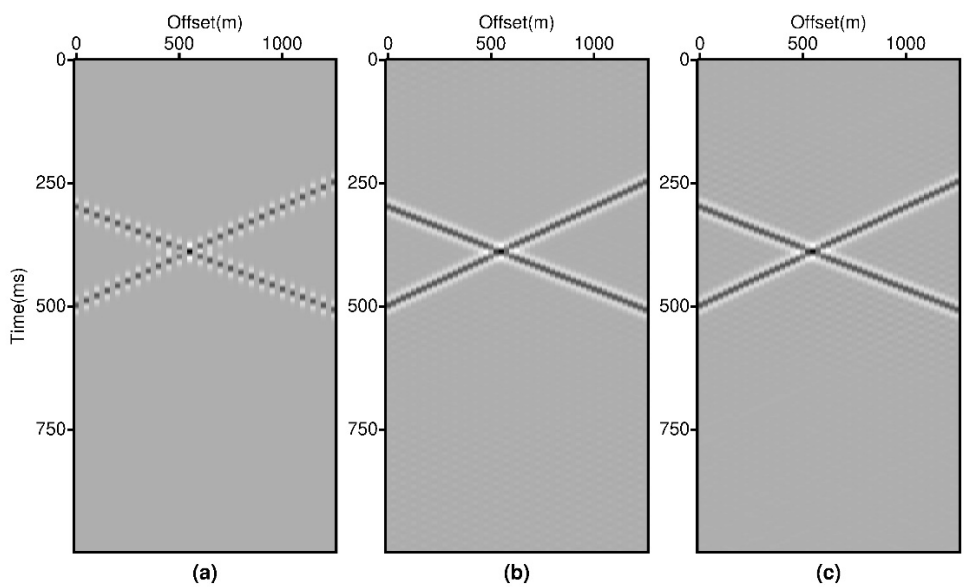


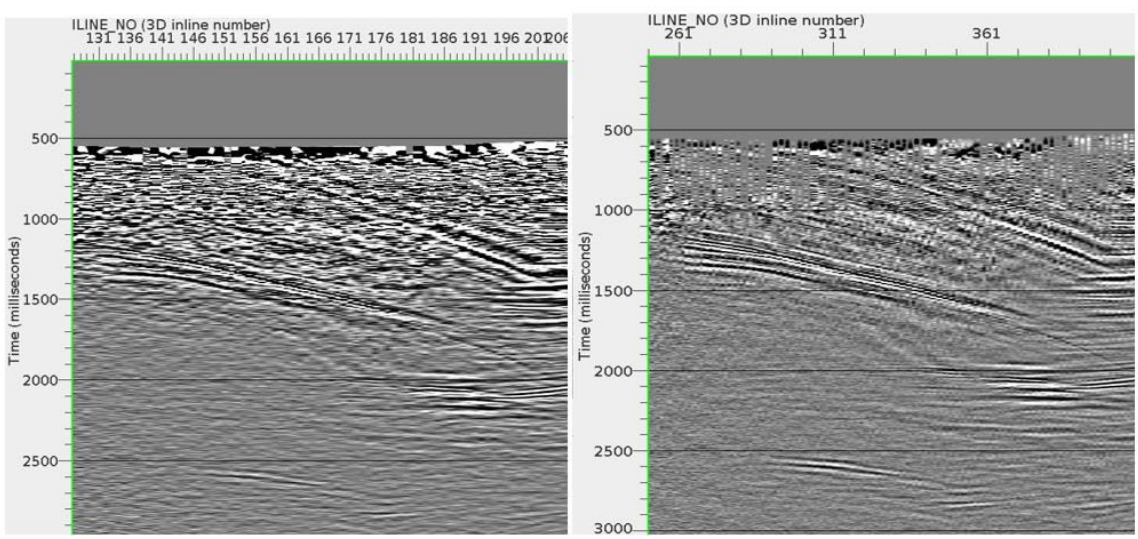

Figure 9: Comparison of anti-aliasing greedy Radon transform and angular weighted MWNI (Ng et al, 2015). (a) Input data. (b) Result of anti-aliasing greedy Radon transform. (c) Result of angular weighted MWNI.

The comparison of the amplitude of Radon models (see Figure 8) shows that conventional greedy Radon transform can really enhance the resolution of the model. Unfortunately it cannot eliminate the unwanted curves related to spatial aliasing. On the other hand, by adaptively stacking amplitude of each moveout parameter from low frequency to high frequency and using the amplitude stack as a preconditioner, the new algorithm can suppress aliasing and provide a stable Radon model for the true signal.

Figure 9 compares the result of the angular weighted MWNI and that of the new greedy Radon method. Careful comparison shows that the new method provides a more stable result than the angular weighted MWNI. In particular, the angular weighted MWNI generates more artifacts than the greedy Radon transform, though they are much weaker than the signal.

## Real data example

We also validate the algorithm with a real 3D dataset. The natural CDP bin spacing of the seismic survey is 25 meter in the inline direction and 50 meter in the crossline direction. Because the geology is higly structured, the stacked image in the crossline direction shows strong aliasing and smearing (see Figure 10a). After rebinning the data onto a 25 m x 25 m CDP grid, we can certainly improve the image resolution as shown in Figure 10b. However there are still two outstanding issues in the result. First of all near offsets are missing in every second CDP, which creates gaps in the stack image in the shallow zone. Second of all the aliasing problem becomes more evident due to lower fold in each CDP gather and higher resolution across CDPs.




(a) (b)

Figure 10: Stacked images with different binning parameters. (a) Stacked image using 25 m x 50 m grid. (b) Stacked image using 25 m x 25 m grid.

To address the two problems, we tested MWNI and the proposed anti-aliasing greedy Radon transform with 4D implementation on this challenging dataset. Figure 11 compares the input gather and the regularized output gathers by the two interpolation methods. Both algorithms

successfully recover missing traces due to the survey limitation. Overall the two results look similary. However the greedy Radon transform seems to provide a cleaner output. In particular some of the linear noise is attenuated.

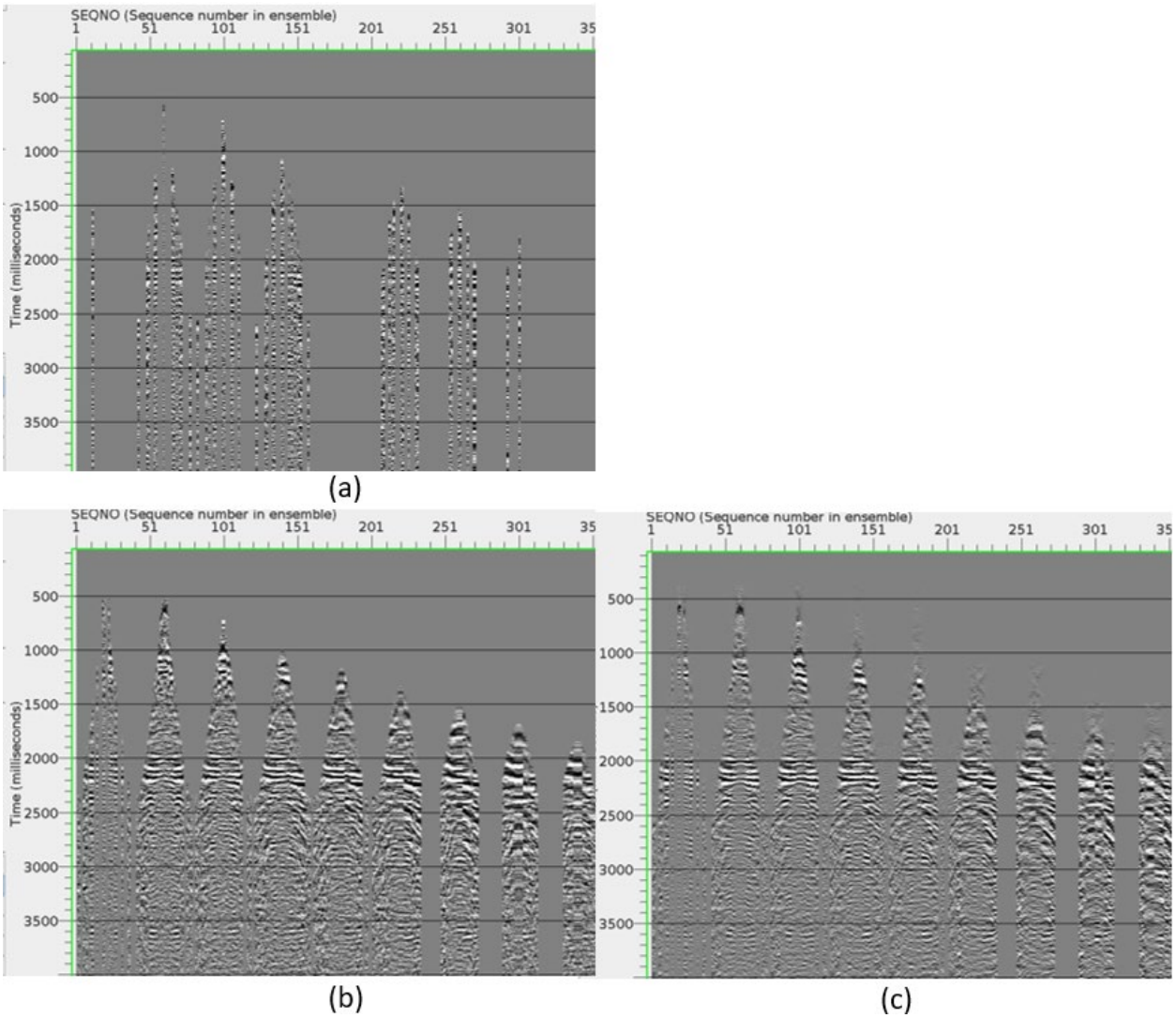



Figure 11: Comparison of input and interpolated gathers with MWNI and greedy Radon transform.(a) Input CDP gather binned based on regular y-offset and x-offset. (b) Regularized CDP gather using MWNI method. (c) Regularized CDP gather using anti-aliasing greedy Radon transform. Both (b) and (c) are sorted in the same order as (a).

The stack comparison (see Figure 11) shows that both MWNI and the greedy Radon transform can fill in the shallow gaps. The signal-to-noise ratio of the stacks increases due to higher CDP fold after interpolation. It is obvious that the greedy Radon transform enhance the continuity of events better than MWNI in this case. The image looks much sharper, particularly in the shallow zone. More importantly, most of the aliasing is suppressed after interpolation. Subtle structures

are revealed after the gaps are filled in and coherent events are properly connected without smearing.

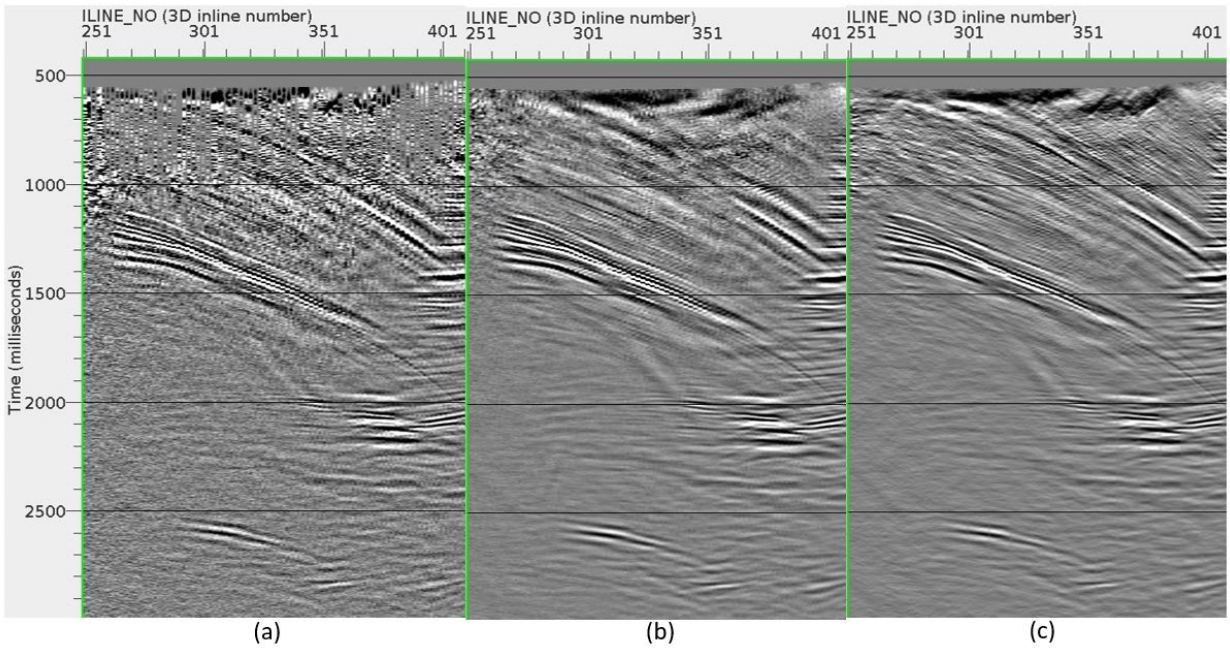



Figure 11: Comparison of stacks before and after interpolation. (a) Input stack. (b) Stack after MWNI. (c) Stack after anti-aliasing greedy Radon transform.

## Conclusions

I have implemented a new interpolation method utilizing multi-dimension greedy Radon transform that can address spatial aliasing in seismic data. Synthetic and real data examples show that the method can provide more stable and sharper image than the conventional MWNI algorithm. By setting the linear moveout range in CDP range, the algorithm can naturally suppress unwanted lienar noise. Therefore we can use the method to perform interpolation and noise attenuation at the same time.

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